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L intersección de ideales es un ideal

Ejercicio 1. Sea $\{I_\lambda\}_{\lambda\in\Lambda}$ una colección de ideales de un anillo R. Demuestra que

$$\bigcap_{\lambda\in\Lambda} I_\lambda \text{ es un ideal de } R.$$

***Solución*:** Comprobemos las propiedades de la `Ideal/`Definición 1ideal.pdf:

(i) $0 \in \bigcap_\lambda I_\lambda$ porque $0 \in I_\lambda$ para todo $\lambda \in \Lambda$.

(ii) Sean $a, b \in \bigcap_\lambda I_\lambda$, entonces $\forall\lambda \in \Lambda : a, b \in I_\lambda$. Como I_λ es un ideal, $\forall\lambda \in \Lambda : a+b \in I_\lambda$ y concluimos que $a + b \in \bigcap_\lambda I_\lambda$.

(iii) Sea $a \in \bigcap_\lambda I_\lambda$ y $r \in R$, entonces $\forall\lambda \in \Lambda : a \in I_\lambda$ y por la propiedad de absorción, $\forall\lambda \in \Lambda : ra \in I_\lambda$. Por tanto, $ra \in \bigcap_\lambda I_\lambda$.

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